

COLLINS CULBERT

Electrical Systems Engineering Student · Embedded Systems & Robotics Control

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SUMMARY

Junior Electrical Systems Engineering Technology student with hands-on experience across closed-loop robotics control, FPGA digital logic, and register-level embedded programming. Led the engineering build of a **10-servo, PID-controlled robotic hand** that earned a **1st Place — Best Presentation award** and finalist standing. Comfortable working end to end: schematic to firmware to live demo.

TECHNICAL SKILLS

Programming	Python, SQL, C/C++ (STM32 bare metal), MicroPython, JavaScript, HTML/CSS, MATLAB
Hardware & Tools	STM32F4xx / Keil uVision, Quartus Prime (FPGA/DE10), KiCad, SolidWorks, oscilloscopes, multimeters, DC motor drivers
Specializations	Closed-loop PID control, I ² C sensor integration, FPGA state machine design, register-level GPIO/timer programming
Software	MS Office Suite, Oracle, HubSpot (basics)

RELEVANT COURSEWORK

ESET 210 - Circuit Analysis · **ESET 269** - Embedded Systems Development in C · **ESET 369** - Embedded Systems Software · **PHYS 207** - Electricity & Magnetism · **MATH 251** - Calculus III

PROJECTS

Robotic Hand — Closed-Loop PID Control

MicroPython, RP2040/ESP32, AS5600, PCA9548A, I²C

- ▶ Co-designed and built a **10-servo DC motor-driven robotic hand** (2 continuous-rotation servos per finger) for the Texas A&M Robotic Hand Challenge.
- ▶ Engineered a **closed-loop PID control architecture** using **AS5600 magnetic encoders** read over I²C through a PCA9548A multiplexer for per-finger position feedback.
- ▶ Built a **calibration workflow** in MicroPython to capture and store named hand poses from live encoder data.
- ▶ Delivered the technical presentation that earned the team the **1st Place — Best Presentation Award** and overall finalist standing.

FPGA Motor Speed Controller

Quartus Prime, DE10 FPGA, Verilog/VHDL, PWM

- ▶ Designed and implemented a **7-state, one-hot Moore state machine** PWM motor speed controller on the DE10 FPGA.
- ▶ Drove a **BCD 7-segment display** output directly from FSM state logic.

STM32F4xx Embedded Systems Lab

C (bare metal), STM32F4xx, Keil uVision

- ▶ Programmed **GPIO peripherals and SysTick timers** at the register level in C with no HAL abstraction.

EDUCATION

B.S., Electrical Systems Engineering Technology (in progress)

Expected Dec. 2028

Texas A&M University, College Station, TX · **Regents Scholar**

LEADERSHIP & INVOLVEMENT

Cohort 2 Member, KCLI Brotherhood

College Station, TX · Sept. 2024 - Present

- ▶ Oversee compliance with organization and chapter safety rules and regulations for **20+ members**.